

PSYCH 238 A&B: BIOPSYCHOLOGY

Fall 2026 St. Olaf College

Section A: RNS 190

Tuesday 11:45 AM-1:10 PM (85 min)
Thursday 12:45-2:05 PM (80 min)

Section B: RNS 190

Tuesday 1:20-2:45 PM (85 min)
Thursday 2:15-3:35 PM (80 min)

Instructor: Dr. Brian Rivera — rivera21@stolaf.edu — Office: RNS 160A

Office Hours: Mondays and Wednesdays 1:00-2:30 PM or by appointment

Credits: 1.00 (IST General Education) — Prereq: PSYCH 125, or BIO 123 and BIO 227

1 ♥ Course Description

This course provides a comprehensive introduction to the biological foundations of behavior using multiple levels of analysis (**molecular, cellular, systems, and behavioral**). Students will critically examine research methodologies in biological psychology while learning how **brain anatomy, physiology, and chemistry** underlie **sensation, perception, memory, and emotion**.

This course emphasizes how biopsychology impacts various areas of psychology and everyday behaviors, developing students' scientific reasoning skills to understand how biological mechanisms contribute to psychological processes including **learning, memory, stress, and sleep**. As an inherently interdisciplinary endeavor, this course satisfies the Integrated Scientific Topics (IST) General Education requirement by integrating scientific content and principles from multiple disciplinary perspectives.

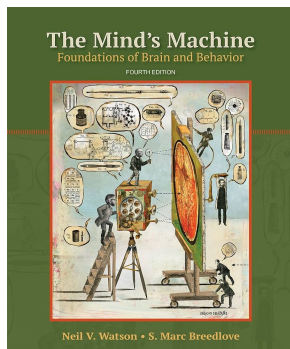
2 ♦ Learning Objectives

By the end of this course, students will be able to:

1. **Describe** the historical foundations of biological psychology and identify key developments in neuroscience.
2. **Identify** major brain structures and explain basic principles of neural communication and neurotransmission.
3. **Explain** how neurobiological processes underlie both normal and abnormal behaviors.
4. **Analyze** primary research articles in biopsychology, evaluating methodology and interpreting findings.
5. **Write** clear, scientifically accurate reports using appropriate biopsychological terminology.
6. **Critically assess** popular media claims about brain and behavior research, distinguishing science from speculation.
7. **Present** and defend research on a brain region orally, answering questions from an instructor and peers.
8. **Become** more informed consumers of media information regarding biopsychology topics.

3 ♣ Required Materials

Textbook: The Mind's Machine: Foundations of Brain and Behavior, **4th edition** (Watson & Breedlove)



Textbook Access: The required textbook is available for checkout at the Rolvaag Library circulation desk.

Additional Resources: As part of this course, you will be expected to use library resources to find scientific articles. If you are not sure how to do this, please ask! I am happy to help, and the reference librarians are excellent resources.

Additional: Moodle access. Bring paper and a pen every day; reflections are handwritten in class. A laptop is only needed for the Moodle quizzes, which you do outside class. The Navigation & Songlines reading (Norris & Harney, 2014) will be available on Moodle.

4 ♠ Important Dates

Key Dates:

- Add: Sep 23rd
- Fall Break: Oct 10th-13th
- Drop: Nov 11th
- Thanksgiving: Nov 25th-29th
- Last day of class: Dec 15th
- Finals: Dec 17th-22nd

5 ♥ Assessment and Grading

Grade Distribution

Grading Scale

Assessment	Points	%
Exams (3)	600	60%
MFBA Project & Poster Session	200	20%
Reading Quizzes (10)	100	10%
Class Participation	70	7%
Reflections (3)	30	3%
Total	1000	100%

Grade	Range
A	930-1000
A-	900-929
B+	870-899
B	830-869
B-	800-829
C+	770-799
C	730-769
C-	700-729
D+	670-699
D	600-669
F	Below 600

Final grades are not rounded. 929 points is an A-, 899 is a B+, and so on.

6 ♦ Reading Schedule

Class	Day	Date	Topic	Reading	Assignments
1	Th	Sep 10	Intro: Course, History & Methods	Syllabus, p. 2-15	
2	T	Sep 15	Ch 1: The Neuron	p. 16-25	
3	Th	Sep 17	Ch 1: Brain Anatomy	p. 25-40	Quiz 1 (due 9/16)
4	T	Sep 22	Ch 2: Action Potentials	p. 54-69	
5	Th	Sep 24	Ch 2: Synaptic Transmission	p. 70-75	Quiz 2 (due 9/23)
6	T	Sep 29	Ch 2: Brain Electrical Activity	p. 76-81	
7	Th	Oct 1	Ch 3: Brain Chemistry	p. 82-87	
8	T	Oct 6	Ch 3: Neurotransmitters	p. 88-93	Quiz 3 (due 10/5)
9	Th	Oct 8	Ch 3: Drugs	p. 93-111	Reflection 1
		Oct 10-13	<i>Fall Break</i>	<i>No Classes</i>	
10	Th	Oct 15	Ch 4: Brain Development	p. 120-130	MFBA Topic (due 10/14)
11	T	Oct 20	Exam 1 Review		
12	Th	Oct 22	EXAM 1: Ch 1-4		
13	T	Oct 27	Ch 5: The Sensorimotor System	p. 142-153	
14	Th	Oct 29	Ch 5: Movement	p. 154-174	
15	T	Nov 3	Ch 6: Hearing & Balance	p. 176-196	Quiz 4 (Ch 5, due 11/2)
16	Th	Nov 5	Ch 7: Vision Pathways	p. 212-230	Quiz 5 (Ch 6), MFBA Biblio (due 11/4)
17	T	Nov 10	Ch 7: Color, What & Where	p. 231-242	
18	Th	Nov 12	Exam 2 Review		Quiz 6 (Ch 7, due 11/11), Reflection 2
19	T	Nov 17	EXAM 2: Ch 5, 6 & 7		
20	Th	Nov 19	Ch 10: Sleep	p. 326-346	MFBA Outline (due 11/18)
21	T	Nov 24	Ch 11: Stress	p. 367-375	Quiz 7 (due 11/23)
		Nov 25-29	<i>Thanksgiving Break</i>	<i>No Classes</i>	
22	T	Dec 1	Ch 13: Memory & Learning	p. 406-432	Quiz 8 (due 11/30)
23	Th	Dec 3	Ch 14: Attention	p. 434-452	Quiz 9 (due 12/2)
24	T	Dec 8	Navigation & Songlines	Norris & Harney 2014	Quiz 10 (due 12/7), Reflection 3
25	Th	Dec 10	MFBA Poster Session I		Report due 12/9
26	T	Dec 15	MFBA Poster Session II		Course Evaluation
	Sat	Dec 19	FINAL EXAM, Section A: 1:00-3:00 PM		Ch 10, 11, 13, 14 & Navigation
	T	Dec 22	FINAL EXAM, Section B: 1:00-3:00 PM		Ch 10, 11, 13, 14 & Navigation

Online assignments are due at 11:59 PM on the date in parentheses. Reflections and the course evaluation happen in class on the day listed.

7 ♣ Assessment Details

7.1 Exams (600 points)

Three noncumulative exams, each worth 200 points. All students must take all three exams. Multiple-choice (40%), short answer (30%), and application questions (30%). **Exam questions will be based on lecture notes as well as the readings, so consistent attendance is important for doing well.** The exact format and what to bring will be shared in class before each exam.

- **Exam 1:** October 22nd (Thursday) - Chapters 1-4
- **Exam 2:** Nov 17th (Tuesday) - Chapters 5, 6 & 7
- **Final:** Section A: Saturday, Dec 19th, 1:00-3:00 PM; Section B: Tuesday, Dec 22nd, 1:00-3:00 PM - Chapters 10, 11, 13, 14 & Navigation

7.2 MFBA Project & Poster Session (200 points)

My Favorite Brain Area (MFBA): Pick one brain region and become the class expert on it. You will investigate its anatomy, physiology, neurochemistry, and research history, plus clinical relevance, and connect it to the wider networks and functions it takes part in. You may work alone or in a group of up to three; groups turn in one report and one poster and share the grade. Groups are set by the topic proposal deadline (Oct 14), and each group member submits a short statement with the report describing what they contributed. Brain areas are claimed on Moodle, first come first served. If two students or groups propose the same area, they present it together as one group; I will not evaluate the same brain area twice.

Report (60 points): 6-8 pages covering the four core domains (anatomical structure, physiological function, neurochemical properties, historical development of research on the region) plus clinical applications and the research methods used to study it. Minimum 10 peer-reviewed sources, a mix of foundational and recent studies. Detailed guidelines and the rubric, including how large or small a brain area can be, will be provided with the topic proposal instructions. Due Dec 9 (11:59 PM).

Poster session (140 points): The last two class meetings (Dec 10 and Dec 15) are a poster session. Half the class presents each day while the other half evaluates. I go station by station; when I reach yours you give me a 5-7 minute presentation of your region and answer questions, in the style of an oral exam. Your classmates evaluate too: each of you is assigned specific stations to visit and grade, using sheets I provide. I score 100 points (content accuracy, depth, ability to answer questions, clarity of the poster); peer evaluations make up the other 40.

Poster format: low tech. Design your poster as PowerPoint slides (or equivalent), print the pages, and tape them to the wall at your station. No poster printing, no screens. Presenting order and station assignments are posted on Moodle before Dec 10 and are fixed. Missing your presenting day is a 0 for the poster session; there is no make-up. Evaluating your assigned stations on the day you are not presenting is required and counts as that day's attendance and participation point.

Milestones:

- Topic proposal: Oct 15th
- Annotated bibliography (10 sources): Nov 5th
- Outline: Nov 19th. I will be available to meet about your project between Nov 19th and Dec 4th
- Report due: Dec 9th
- Poster session: Dec 10th and Dec 15th (your presenting day is assigned on Moodle)

7.3 Reflections (30 points)

Three short reflections, written by hand in class during the last 20 minutes of the meetings marked in the schedule (Oct 8, Nov 12, Dec 8). Open notes and open book. The prompt is given in class and is the same each time: one idea from this block that changed how you think about something in your life; one claim about the brain you have seen in the media that you now read differently, and why; and one question you still have. Each reflection is worth 10 points (10 full, 7 thin, 4 off-prompt, 0 missing). Reflections cannot be made up outside class except with a documented absence.

7.4 Class Participation (70 points)

Participation is worth 70 points in three parts: Inquisitor Days (30), Participation Points (30), and Attendance (10).

Inquisitor Days (30 points). Each class has three or four Inquisitors. You will be an Inquisitor three times; your dates are posted on Moodle. On your day, submit one question about that day's reading on Moodle by 11:59 PM the night before (5 pts), then ask it in class (5 pts). I read the questions before class and use them to decide what to explain and how.

What makes a good question: it should be something you actually want answered, it should be about the reading, and it should be worth the whole class hearing. Two kinds work well. The first is a question the reading left you with: something you didn't follow, or that seemed to contradict something earlier in the course. The second is a why or how question about a mechanism the textbook describes but doesn't fully explain: why does the axon need myelin if ions already flow, or how does a drug that blocks reuptake end up changing mood days later.

What doesn't work: a question that only asks me to repeat a part of the reading ("can you go over the parts of the neuron again"); a hypothetical ("what if the hippocampus were missing"), since those are easy to invent and take us away from the mechanism; and a question that is really about a different course. A very specific question about a molecular pathway or a clinical detail the reading never touches is not useful to the class, even if it's a good question in itself. Stay within reach of the chapter.

Submitted question (5 points):

Pts	Description
5	Asks why or how about a mechanism and opens a door: answering it takes the class somewhere the reading doesn't go, while staying tied to it.
4	About the reading, asks why or how about a mechanism, and worth the class hearing.
3	About the reading and gives me something to explain, but only asks for a repeat or a definition.
2	Hypothetical, or drifts outside the chapter.
1	Submitted on time but vague or unrelated to the day's reading.
0	Not submitted by 11:59 PM the night before.

Asking it in class (5 points): when I call on you, be ready to say where in the reading the question came from and what you were thinking when you wrote it, not just read it aloud. 5 if asked with that context, 3 if just read, 0 if absent or silent.

Participation Points (30 points). A participation point is worth 3 points, at most one per class day, so you need ten over the semester for full credit. To earn one, the contribution has to be substantial: an answer with reasoning behind it, or a question that shows you've thought about the material. Blurting out a guess doesn't count. You earn one by:

1. Answering a question I pose to the class. Right or wrong, as long as you can say why you think so.

2. Asking a question about the material on a day you are not an Inquisitor.
3. Talking with me about the material after class.
4. Emailing me a relevant question about the content, or something new you've learned that connects to it (not about the course, grades, or logistics).
5. On the rare days we do a small group activity, summarizing your group's work for the class.
6. Completing the course evaluation during the last class (Dec 15) and emailing me that you have done so. Evaluations are anonymous; your email is the only record I have. Counts as that day's participation point.

Raising your hand and being wrong is hard. It's also where the conversation starts, and that conversation is how you learn the material. A wrong answer with reasoning behind it earns the point.

Attendance (10 points). Three unexcused absences are free; each one after that costs 2 points. See the Attendance Policy below.

7.5 Reading Quizzes (100 points)

Ten online quizzes (10 pts each) via Moodle. Each quiz opens at the end of the previous class and is due at 11:59 PM the night before the class listed in the schedule. 10 questions, 15 minutes. Open book open notes. **No quiz is dropped.**

Where your grades are: everything is posted in the Moodle gradebook. Participation Points and Inquisitor scores are tallied by hand and entered around Fall Break and again after Thanksgiving; ask me at any time where you stand.

8 ♠ Course Policies

8.1 A Typical Class

About 80% of each meeting is lecture. The day's Inquisitors ask their questions along the way and I pose questions to the room; that is where most Participation Points are earned. After a short break we finish with an activity or discussion, and the last few minutes are housekeeping: announcements, upcoming deadlines, and questions about the course. The lecture assumes you have done the reading.

Sections: you are enrolled in one section and attend that section. Sitting in on the other section does not count as attendance, as an Inquisitor day, or as a poster session day.

8.2 Academic Integrity & Plagiarism

Plagiarism is not only taking someone else's ideas and presenting them as your own without proper citation, it is also changing a word or two and using citations inappropriately.

I take academic dishonesty very seriously and any instances will result in, at minimum, a failing grade for that assignment and possibly a failing grade in the course. For complete details on St. Olaf's academic integrity policy, see: <https://wp.stolaf.edu/thebook/plagiarism/>

8.3 AI Tool Usage Policy

Student process with limited, specific AI usage. In this course, AI is allowed in a limited way. Learning how to use AI for certain tasks can be useful, and you may need to know how to use it post-

graduation. Therefore, this course challenges you to use AI ethically and responsibly while maintaining your own integrity, creativity, and critical thinking, your best features as a human being.

You should think of AI as a utility that aids your learning process, but an imperfect one. AI can assist you in fine-tuning your original ideas, but all initial drafting should be your own; AI should not “think” or “create” content for you. Always assume that what AI gives you might be wrong, containing incorrect, incomplete, or biased information. The more you substitute AI output for your own ideas and work, the more you risk circulating incorrect information or perpetuating bias.

This means you are responsible for:

- Critically engaging with readings, lectures, discussions, and other materials.
- Using this learning to develop approaches, research, and insights that become the submitted work for the course.
- Composing first drafts of assignments using your intellect and creativity.
- Revising and iterating assignments through feedback from peers and your professor.
- Ensuring that any AI practices adhere to the uses and responsibilities below.

You may use AI to:

- Get started by brainstorming initial questions or directions prior to research or drafting.
- Help you reflect on concepts, strategies, and areas of concentration for revision, without revising for you.
- Check for grammar and style, but skeptically.

If you use AI, you should:

- Provide documentation tracking how AI assisted in helping you narrowly fine-tune your process.
- Verify and cite all information (numbers, data, quotations, sources, claims, etc.) per standards outlined in class.
- Avoid using AI on quizzes or exams or to substitute for the drafting and revision of assignment content.
- Think carefully about the ethics of AI and work to minimize its harms.

The value of an education is the cultivation of your mind. Using LLM outputs to substitute for critical thinking may stunt your learning and prevent you obtaining skills that will distinguish you after graduation in professional and civic life. So, don’t let a machine think for you; that’s boring and unethical. And if Oles are known for anything, it’s neither of those.

By enrolling in this course, you pledge to uphold these policies, conducting your work according to your professor’s guidelines. You recognize that failure to do so means that you have likely cheated, plagiarized, or falsified information in violation of St. Olaf’s Academic Integrity policy and Honor Code.

In this course specifically: Inquisitor questions may be AI-assisted, but the question you submit must be one you understand and want answered, and when I call on you, you will explain where it came from in your own words. Reflections are handwritten in class.

Instructor’s AI Use: I use AI extensively for course planning, research, coding, and formatting. I’ve built a library of digital diagrams using AI that I will be sharing throughout the semester. I am responsible for verifying all AI-generated content for accuracy before sharing.

Important Notes: This policy applies to this course only. Other professors may have different policies, and we will adapt to any college-wide policy changes.

8.4 Attendance Policy

Regular attendance and punctuality are essential. I expect everyone to be present and ready to participate every day, and attendance will be incorporated into your participation grade.

- **Allowed absences:** 3 unexcused absences maximum. Each additional unexcused absence costs 2 points from the attendance portion of your participation grade.
- **Tardiness:** Please arrive on time. It is disruptive and rude to arrive late. Try to use facilities before class begins.
- **Excused absences:** Whether an absence is excused is my call. I do not require a doctor's note, and I will take you at your word. That trust is the whole system: misrepresenting the reason for an absence, a late assignment, or a missed exam is academic dishonesty and will be handled under the Academic Integrity policy above, which means at minimum a failing grade for the work involved.
- **Communication:** Let me know about anticipated absences sooner rather than later. Life happens.

8.5 Technology Policy

- **Cell phones:** Must be silenced and put away during class. Texting or posting during class is rude and disruptive.
- **Laptops:** Note-taking on computers is acceptable, but please refrain from surfing, social media, and emailing during class. This affects not only you but also your neighbors.
- **During exams:** No electronic devices permitted.

8.6 Late Work & Make-up Policy

- **Quizzes:** Cannot be made up. Each quiz is open for several days; plan for it.
- **Inquisitor questions:** Not accepted late; the submitted-question points are lost, but you can still earn the in-class points by asking. If you cannot make your Inquisitor day, trade with a classmate and email me before the submission deadline. A no-show on your day is 0 for that day.
- **Poster session:** No make-up. Missing your presenting day is 0 for the poster session.
- **Reflections:** Written in class only. If you will miss a reflection day, email me before class starts; a make-up is at my discretion and the honesty rule in the Attendance Policy applies.
- **Major assignments:** Late work requires prior arrangements. Contact me *before* the deadline if you anticipate problems.
- **Exams:** If you cannot take an exam, email me before the exam starts. A make-up is at my discretion and the honesty rule in the Attendance Policy applies. Do not assume you will be able to take a make-up exam.
- **Communication is key:** If you know you'll be unable to complete work on time, talk to me about it ahead of time!

8.7 Classroom Etiquette & Expectations

Respectful Learning Environment:

- Only one person should speak at a time - side conversations are distracting
- Respect the ideas and contributions of your classmates

Academic Commitment: Students expected to devote minimum 3 hours outside class per hour in class.

My Responsibilities:

- Be reasonably available to answer questions and help you formulate questions.
- Help you explore your interests in this fascinating material.
- Grade assignments fairly and in a timely fashion.
- Respect your ideas and contributions.

9 ♡ Support & Success Resources

9.1 Disability Accommodations

If you have a documented disability for which accommodations may be required in this class, please contact the Academic Support Center as soon as possible to discuss accommodations. If you have already arranged accommodations through Student Accessibility Services, please arrange for submission of your accommodation letter within the first two weeks of class.

Accommodations will only be provided after the letter is submitted with sufficient lead-time for me to arrange testing or other accommodations. Although I will receive the letter electronically, I expect you to initiate a conversation with me about the accommodations. Please do this by the end of the second week so that extended-time testing can be scheduled for the exams, including the Saturday final for Section A.

Contact: Academic Support Center, Buntrock 108, 507-786-3288

9.2 Additional Support & Resources

Getting Help: Should you encounter difficulties with course material, please contact me during office hours, via email, or after class. I am committed to supporting your academic success.

Communication: Course announcements go out on Moodle. I answer email within one business day; I do not answer on weekends, so a question about a Monday deadline should reach me by Friday.

Supplemental Instruction (SI)

Supplemental Instruction (SI) is a program that assists students in learning and retaining course information, is statistically proven to increase student grades by one letter grade, and enhances course retention and persistence.

Psych 238 A/B – Rivera

SI Leader: Mary Cate Pugh

Location: TOH 108

Schedule:

- Sundays: 5:00–6:00 PM
- Mondays: 7:00–8:00 PM
- Wednesdays: 7:00–8:00 PM

Students are strongly encouraged to attend these free study sessions to reinforce course concepts through collaborative learning and peer support.

Study Resources:

- Form study groups with classmates to discuss material
- Use library resources for research - ask reference librarians for help
- Visit the Writing Desk (Tomson 157) for help with the MFBA report at any stage: <https://wp.stolaf.edu/academic-support/writing-help/>
- Research librarians can help you find and cite sources for the MFBA: <https://www.stolaf.edu/library/research/appointments/>

Academic Support: Take advantage of all the resources St. Olaf provides. This course requires consistent effort, but with the right support and strategies, you can succeed!

9.3 College Statements

Inclusivity: In keeping with St. Olaf College's mission statement, this class strives to be an inclusive learning community, respecting those of differing backgrounds and beliefs. As a community, we aim to be respectful to all citizens in this class, regardless of race, ethnicity, nationality, religion, gender, sexual orientation, or disability.

Mental health: If you are experiencing mental health symptoms or issues, multiple resources are available to you on campus, including the Dean of Students office (<https://wp.stolaf.edu/deanofstudents/>), the Counseling Center (<https://wp.stolaf.edu/counseling-center/>), and TimelyCare (<https://wp.stolaf.edu/timelycare/>). Please do not hesitate to come see me if you need assistance connecting with the support you need.

Required referrals: You are welcome to speak to me about any issues that arise for you in the context of this course, including any circumstances outside of class that may be affecting your classroom experience or academic performance. However, please be aware that I am required to refer to the college's Director of Equal Opportunity any information students share about experiences of sexual harassment, sexual assault, or dating violence. I am also expected to share any disclosures of incidents of bias, harassment, or discrimination to the Bias Response Team. Confidential resources that do not have this obligation include the Counseling Center, College Ministry (<https://wp.stolaf.edu/ministry/>), the Sexual Assault Resource Network (<https://pages.stolaf.edu/sarn/>), Health Services (<https://wp.stolaf.edu/health/>), and TimelyCare. More information: <https://wp.stolaf.edu/title-ix/support/> and <https://wp.stolaf.edu/community-belonging/>.

9.4 Final Exam Policy

The final exam time and date are not negotiable; all students are required to be present on the date and time specified for the course. College policy prohibits me from making any special arrangements except in dire circumstances.

If you have a serious & documented medical or other emergency, you must present your case to the chair of the psychology department, who will evaluate your request for an alternate final date/time. **This is the only exception to the final exam schedule.**

Syllabus subject to change with advance notice. Updates announced in class and on Moodle. All dates and topics are subject to change! We might need to slow down or decide to explore a different topic. Exam dates will only be changed by agreement of the entire class. Updated 9/9/2026

